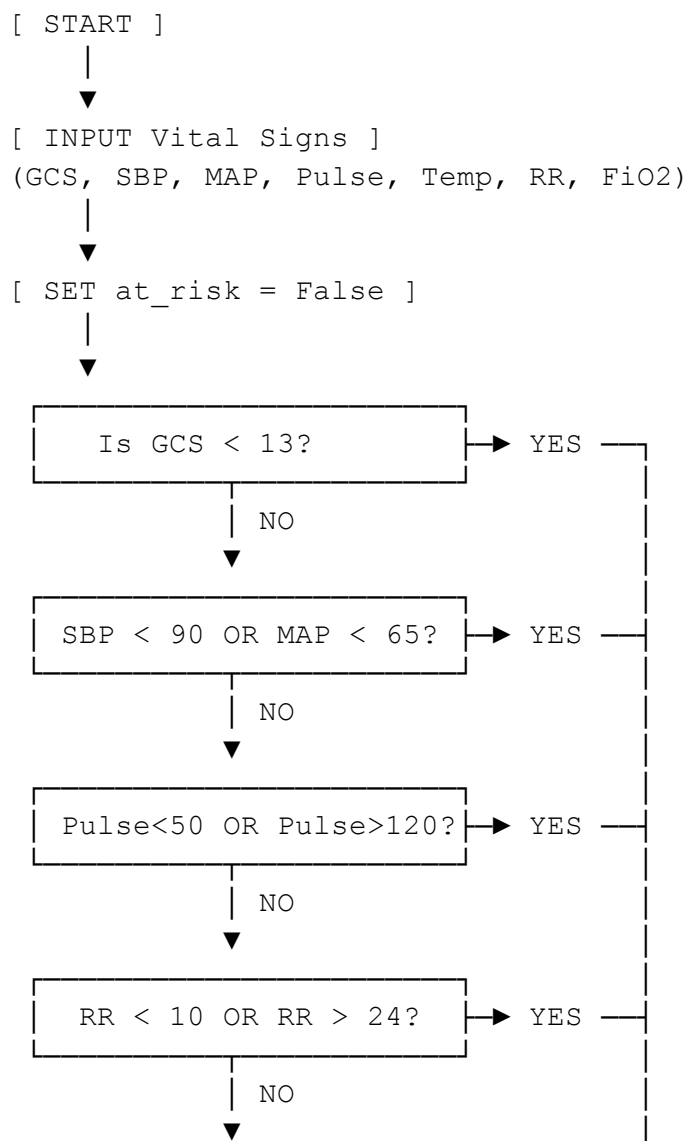


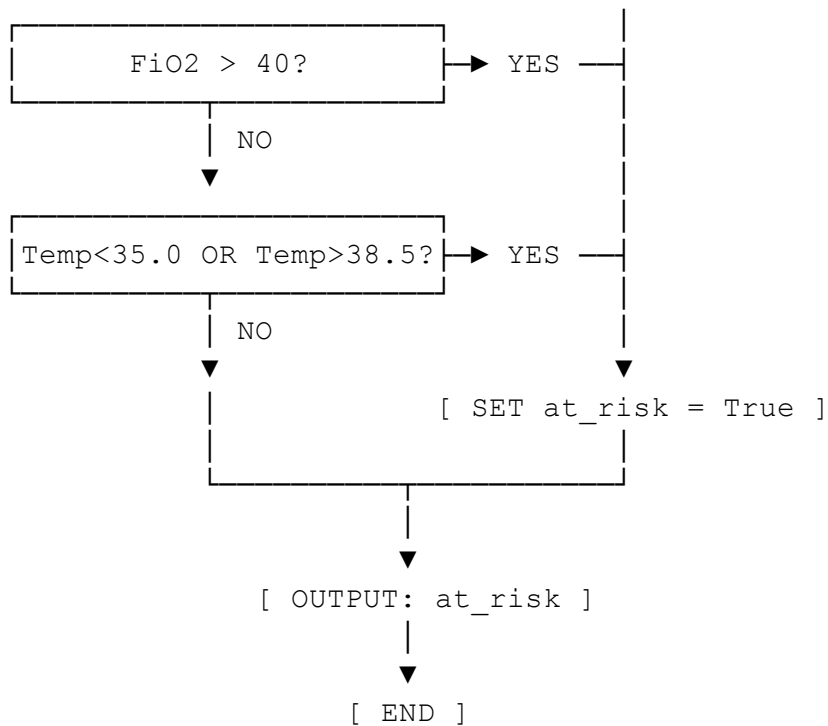
Mercer ED Triage Pseudocode Model

Approach

A simple rule-based system was developed using triage-stage vital signs available within 3-5 minutes of patient arrival. Each variable was mapped to a key physiological system (neurological, cardiovascular, respiratory, and systemic) and clinically meaningful thresholds were applied based on standard emergency triage practice. A patient is flagged as at risk if any single threshold is breached, prioritising early detection of deterioration over specificity.

Pseudocode





Justification

This rule-based system is designed for the triage stage of Mercer General ED, where clinicians have 3-5 minutes and only access to vital signs and brief clinical information. It uses early physiological markers across neurological (GCS), cardiovascular (SBP, MAP, pulse), respiratory (RR, FiO₂), and systemic (temperature) systems to identify patients at risk of deterioration.

The thresholds reflect common emergency triage concerns such as shock, hypoxia, altered consciousness, and sepsis. The goal is not diagnosis, but early identification of potentially unstable patients who may require urgent assessment (ESI 1-2). The rules prioritise sensitivity to ensure high-risk patients are not missed in a time-constrained environment.